

Day 20 of training

useState Hook in React

Definition:

`useState` is a React Hook that allows functional components to have state (data that changes over time). Before hooks, only class components could have state.

Syntax:

```
const [stateVariable, setStateFunction] = useState(initialValue);
```

- **stateVariable** → current state value
- **setStateFunction** → function to update the state
- **initialValue** → starting value of the state

Key Points:

1. `useState` can store numbers, strings, arrays, objects, or even boolean values.
2. Updating state re-renders the component automatically.
3. Always use the setter function to update state; do not modify state directly.
4. Can use multiple `useState` hooks in a single component.

Basic Example (Counter):

```
import React, { useState } from "react";

function Counter() {

  const [count, setCount] = useState(0);

  return (

    <div>

      <h1>Count: {count}</h1>

      <button onClick={() => setCount(count + 1)}>Increment</button>

      <button onClick={() => setCount(count - 1)}>Decrement</button>

    </div>

  );
}
```

}

For usestate in react we need to import **import React, { useState } from "react";** to access ustate hook

Advantages of useState:

1. Adds dynamic behavior to functional components.
2. Makes UI reactive and interactive.
3. Allows multiple states in a single component.
4. Keeps code clean and simple compared to class components.
5. Works with all data types: numbers, strings, arrays, objects, booleans.